MIL-T-83303C (USAF)
9 SEPTEMBER 1988
SUPERSEDING
MIL-T-83303B(USAF)
28 JANUARY 1975

MILITARY SPECIFICATION

TRUCK, FIREFIGHTING, AIRFIELD RAMP, 4X4, A/S32P-20, TWIN AGENT, SKID MOUNTED

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one type of diesel engine driven, four wheel drive, dual rear wheel, twin agent fire truck with utility body consisting of a skid mounted 400 pound minimum capacity dry chemical fire extinguisher unit and a 500 pound minimum capacity bromochlorodiflouromethane (Halon 1211) fire extinguisher unit.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the soliciation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-F-800	-	Fuel Oil, Diesel
W-B-131	-	Battery, Storage, (Vehicular, Ignition, Lighting and Starting).

MILITARY

MIL-P-514	-	Plates, Identification, Instructions and Marking, Blank
MIL-T-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-C-7580	-	Coupling, Fire Hose, Reusable, High Pressure

Benefical comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WARNER ROBINS ALC/MMEDTB, Robins AFB GA 31098-5609 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or letter.

AMSC FSC 4210

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MIL-P-23377	-	Primer Coating, Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-B-38741	-	Bromochloroditluoromethane, Technical
MIL-C-46168	-	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-C-53039	-	Coating, Aliphatic Polyurethane, Single Component Chemical Agent Resistant
MIL-T-83133	-	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-C-83286	-	Coating, Urethane Aliphatic Isocyanate, for Aerospace Applications
MIL-W-83420	-	Wire Rope, Flexible, for Aircraft Control
MS28889	-	Valve, Air, High Pressure Charging
MS51335	-	Pintle Assembly, Towing, 18,000 Pound Capacity, Manual Release

STANDARDS

FEFERAL

FED-STD-595	-	Colors

MILITARY

MIL-STD-12	-	Abbreviations for Use on Drawings, and on Specifications, Standards and Technical Documents
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-461	-	Electromagnetic Emission and Susceptability Requirements for the Control of Electromagnetic Interference
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-721	-	Definition of Terms for Reliability and Maintainability
MIL-STD-781		Reliability Design Qualification and Production Acceptance Test: Exponential Distribution
MIL-STD-808	-	Finishes, Materials and Processes for Corrosion Prevention and Control in Support Equipment
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	-	Test Reports, Preparation of
MIL-STD-889	-	Dissimilar Metals
MIL-STD-970	-	Standards and Specifications, Order of Precedence, for the Selection of
MIL-STD-1223	-	Nontactical Wheeled Vehicle Treatment, Painting, Identification Marking, and Data Plate Standards
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1595	-	Qualification of Aircraft, Missile and Aerospace Fusion Welders
MIL-STD-1791	-	Designing for Internal Aerial Delivery in Fixed Wing Aircraft

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on the date of invitation for bids or request for proposals shall apply.

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Vessel Code, Section VIII Unfired Pressure Vessels

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulations
Federal Motor Vehicle Safety Standards

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington DC 20591.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 414	-	Aircraft Rescue and Fire Fighting Vehicles
NFPA 1931	-	Fire Department Ground Ladders

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J201	-	In Service Brake Performance Test Procedure Passenger Car and Light Duty Truck
J293	-	Vehicle Grade Parking Performance Requirements
J336a	-	Sound Level for Truck Cab Interior
J360	-	Motor Vehicl Grade Parking Performance Test Code
J534	-	Lubrication Fittings
J992b	-	Brake System Performance Requirements - Truck, Bus and Combination of Vehicles
J994	-	Alarm - Back Up - Electric - Performance, Test and Application
J986	-	Sound Level for Passenger Cars and Light Trucks
J1349	-	Engine Power Test Code - Spark Ignition and Diesel

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale PA 15096)

THE TIRE AND RIM ASSOCIATION

Yearbook

(Application for copies should be addressed to The Tire and Rim Association Incorporated, 3200 West Market Street, Akron OH 44313.)

UNDERWRITER LABORATORIES, INCORPORATED

UL 92	-	Standard for Fire Extinguisher and Booster Hose
UL 711	-	Classification Rating and Fire Testing
UL 1093	-	Halongenated Agent Fire Extinguishers

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(Application for copies should be addressed to Underwriters Laboratories, Incorporated, Publications Stock, 333 Pfingsten Road, Northbrook IL 60062.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detailed specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2, 6.2.1 and 6.2.2).

3.2 Design.

3.2.1 Materials. Materials shall be as specified herein. Materials not specifically covered by this specification or referenced document shall be suitable in every respect for this type equipment. Except as otherwise specified, wood shall not be used.

3.2.1.1 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip crack or scale with age or extremes of climatic conditions shall not be used.

3.2.1.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.2.2 Design and construction. The vehicle shall be constructed so that parts do not work loose in service and shall withstand the loads, shocks, vibrations and other conditions incident to operation, shipping and storage with minimum loss of time for maintenance, repair and servicing. The design of the vehicle shall ensure safe, efficient and economical operation. All assemblies, controls and installed equipment shall be located so that there is no adverse interference with each other, or with the operation, and shall be readily accessible for maintenance, operation, removal and replacement. The vehicle shall comply with all applicable Federal Motor Vehicle Safety Standards, Environmental Protection Agency Regulations, Federal Motor Carrier Safety Regulations, National Fire Protection Association and MIL-STD-1472.

3.2.2.1 Reliability. The vehicle, including all components, shall have a Mean-Time-Between-Failures (MTBF) of not less than 100 hours demonstrated in accordance with Test Plan XVIIC of MIL-STD-781. A failure is defined as the event, or inoperable state, in which any item, or part of an item, does not, or would not, perform as specified. Definitions or reliability and terms not included herein shall be as found in MIL-STD-721 and MIL-STD-781. In the event of a conflict between these documents, MIL-STD-721 shall take precedence.

3.2.2.2 Maintainability. The vehicle shall be designed and constructed to provide the following:

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a. A minimum number of parts consistent with the performance required herein.

b. A minimum amount of time for assembling, disassembling, locating trouble sources and maintaining the vehicle. Where practical, components shall be located for rapid inspection and recognition of potential failure.

c. Ease of adjusting, servicing and replacing parts and components.

d. Use of readily available standard tools and equipment for maintenance.

e. Maintenance with a minimum number of tools.

f. Mean corrective maintenance downtime of no greater than one hour.

g. Maximum corrective maintenance downtime of no greater than four hours.

h. Maximum total downtime to perform all daily scheduled maintenance and servicing tasks of no greater than 35 minutes.

The vehicle shall be designed so that one mechanic, using common hand tools, can perform the following maintenance operations within the time specified:

(1) Remove, replace and adjust all engine driven belts - 45 minutes.

(2) Remove and replace alternator - 30 minutes.

(3) Remove and replace engine coolant system hoses - 30 minutes.

(4) Drain engine lubricating oil, remove and replace oil filter element(s) and refill crankcase - 1 hour.

(5) Remove and replace fuel filter element(s) - 30 minutes.

(6) Disconnect battery cables, remove and replace batteries and reconnect battery cables - 30 minutes.

(7) Drain torque converter oil and transmission oil, remove and replace all filter elements and strainers and refill converter and transmission - 1 hour.

(8) Remove and replace starter - 1 hour.

(9) Bleed and adjust brakes and fill master cylinder two mechanics - 45 minutes.

3.2.2.3 Foolproofness. Where improper installation of an item could cause a malfunction, an asymmetric mounting system shall be provided, where practical, to ensure proper mounting of the item.

3.2.2.4 Fasting devices. All screws, bolts, nuts, pins and other fastening devices shall be properly designed, manufactured and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers or nylon patches shall not be used for this purpose, except on the commercial chassis. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter.

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3.2.2.5 Servicing provisions. Drains, grease fittings, lines and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

3.2.2.6 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordance with SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.2.2.7 Component protection. All space in which work is performed during operation, service and maintenance of the vehicle shall be free of hazardous protrusions, sharp edges or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.2.2.8 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached so they cannot become separated from the vehicle.

3.2.3 Environmental conditions. The vehicle shall be capable of satisfactory storage and operation under the following environmental conditions:

a. Full-time exposure to temperatures ranging from -40 degrees F to +125 degree F for operation; -60 degrees F to +160 degrees F for storage.

b. Full-time exposure to relative humidities up to 100 percent.

c. Part-time (five percent of the time) exposure to rain; droplets of size 2.0 to 4.5 mm falling at a rate of 5.2 inches per hour.

d. Part-time (five percent of the time) exposure to salt fog; five percent solution (five parts by weight of salt in 95 parts by weight of water) with fallout of 3.0 ml/80cm2/hour.

e. Part-time (five percent of the time) exposure to blowing dust; velocity 1750 ft/min, concentration 0.5 g/ft3.

f. Part-time (five percent of the time) exposure to blowing sand; velocity 5700 ft/min, concentration 0.0773 g/ft3.

g. Part-time (five percent of the time) exposure to freezing rain/icing; accumulation of 6MM of ice followed by an accumulation of 13MM of ice.

3.2.4 Weight distribution. The weight distribution of the vehicle in any loading condition shall not be unbalanced or improperly distributed. The difference in load between the left and right wheels of any axle shall not exceed 2.5 percent of the total load on the axle.

3.2.5 Body lean. The vehicle shall not lean to either side more than one perce of the body width, in either the fully loaded or the unloaded condition. Body lean

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measurements shall be made at the outermost portion of the vehicle (excluding mirrors, mirror support brackets, lights or assymetrical shapes) directly above each axle; the difference in body height at each axle shall not exceed one percent of the body width at that location.

3.2.6 Electromagnetic interference. The vehicle shall comply with the requirements of MIL-STD-461, Class C1, Group II for electromagnetic interference, except that the limits for UM03 (Figure 8-1) shall be relaxed by 20 dB.

3.2.7 Sound levels.

3.2.7.1 Exterior sound level. The maximum A-weighted sound level produced by the vehicle shall not exceed 88 dBA when measured in accordance with SAE J986. An additional 2 dB allowance over the sound level limit shall be permitted for production units to provide for variations in test site, temperature gradients, test equipment and inherent differences in nominally identical units.

3.2.7.2 Interior sound level. The maximum interior sound level of the cab shall not exceed 84 dBA when measured in accordance with SAE J336a. The reading shall be the maximum A-weighted sound level.

3.2.8 Air transportability. The vehicle with the skid mounted and extinguishing system fully charged (agent tanks and nitrogen cylinders), shall be air transportable in accordance with MIL-A-1791 and AFSC DH1-11 in types C-130, C-141 and C-5 Aircraft. The chassis and skid unit shall also be air transportable as seperate units.

3.2.8.1 Tiedowns. The vehicle shall be equipped with sufficient tiedown points so located that it can be restrained in all directions during air transport. The attachment points shall be designed for use with attachment in the aircraft as described in AFSC DH1-11. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other structure. The tiedown points shall have a minimum clear opening of three inches to allow attachment of a chain with a grabhook.

3.3 Performance.

3.3.1 Mobility. The fully loaded and equipped vehicle shall:

a. Maintain a speed of 70 miles per hour on a level paved road.

b. Accelerate from a stop to a speed of 65 miles per hour within 30 seconds on a level paved road.

c. Ascend and descend a 50 percent grade without spillage of any fluid.

d. Negotiate a 20 percent side slope in each direction.

e. Negotiate, in each direction, a ramp with a 30.6 percent minimum slope connecting two horizontal surfaces. The ramp length shall be at least 1.2 times the wheelbase of the vehicle. The vehicle shall be capable of being driven forward or backward from one horizontal surface across the ramp crest, ascend or descend the ramp, cross the ramp crest and proceed on the

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other horizontal surface without any portion of the vehicle, other than the tires, contacting any of the three surfaces.

f. Traverse graded gravel roads at a speed of 30 miles per hour.

g. Traverse Belgian Block at a speed of 20 miles per hour.

h. Traverse cross country terrain at a speed of 20 miles per hour. MIL-T-83303C

i. Comply with all service brake performance requirements of SAE J201.

j. Comply with all parking brake performance requirements of SAE J293, except that the required grade shall be 20.4 percent.

k. Negotiate eight inch high blocks with diagonally opposite wheels of the vehicle. 1. Traverse, in one pass at low speed, terrain consisting of unflooded fine grained dry to moist soils having a minimum cone index rating in the critical layer of no greater than 20 for fine grained soils and a minimum cone index rating of 50 for coarse grained soils. The vehicle cone index rating shall be measured in accordance with Waterways Experimental Station Mobility Index Method.

m. Maneuver and negotiate sharp turns and curves at an average speed of 40 miles per hour without adverse handling or roadability characteristics.

3.3.2 Halon 1211 agent discharge. The Halon 1211 extinguisher system shall exhibit the following performance characteristics when discharging bromochlorodiflouromethane conforming to at normal operating perssure through 150 feet of one inch inside diameter booster hose:

a. Discharge rate of flow between 5 to 5 1/2 pounds per second (PPS).

b. Extinguish three 600 square foot liquid fuel pan fires in accordance with UL 711, without recharge of agent or nitrogen.

c. Effective discharge of not less than 95 percent of the agent capacity without pressure pulsations or interruptions to flow.

3.3.3 Dry chemical agent discharge. The dry chemical extinguisher unit shall exhibit the following performance characteristics when discharging potassium bicarbonate base dry chemical conforming to O-D-1407 at normal operating pressures through 150 feet of one inch inside diameter booster hose:

a. Discharge rate of between 6 1/2 PPS and 8 PPS when the handline is completely withdrawn from the hose reel (see item d below).

b. Effective conical discharge pattern with a horizontal discharge range of not less than 40 feet without assist from downwind conditions.

c. Effective discharge of not less than 95 percent of the agent capacity without pressure pulsations or interruptions to flow.

d. Discharge rate reduction not exceeding 50 percent when all except five feet of the handline is wound on the hose reel.

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e. Extinguish three 600 square foot liquid fuel pan fires in accordance with UL 711, without recharge of agent or nitrogen. MIL-T-83303C

3.4 Chassis. The chassis shall be the manufacturer's standard commercial item, except as modified to comply with the requirements specified herein. The chassis shall be equipped with four wheel drive, dual rear wheels, and have a minimum gross vehicle weight (GVW) rating of 11,000 pounds for continuous operation.

3.4.1 Powertrain.

3.4.1.1 Engine. The truck shall be equipped with a liquid cooled diesel engine with a minimum of six cylinders. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel in accordance with VV-F-800 and and Grade JP-8 fuel in accordance with MIL-T-83133. The engine shall also operate satisfactorily on Grades JP-4 and JP-5 Turbine fuel in accordance with MIL-T-5624. When operating on Grades JP-4 and JP-5 type fuels, a minimum horsepower degradation is expected but no greater than 25 percent loss is acceptable.

3.4.1.1.1 Engine cooling system. The manufacturer's higest capacity cooling system shall be provided.

3.4.1.1.2 Engine starting system. The engine shall start within 15 seconds in any ambient temperature within the required operating range of the vehicle. Fluid starting aids, glow plugs and heat from the winterization system may be used to facilitate engine starting under the following conditions:

TEMPERATURE RANGE (DEGREES F)	STARTING AIDS PERMITTED
40 through 125	None
1 through 39	Fluid starting aids
-40 through 0	Fluid starting aids, glow plugs and heat from the winterization system

3.4.1.1.3 Exhaust system. The exhaust system shall be designed to expel exhaust gases in a manner that is not hazardous to operators or mechanics; the outlet shall not be directed toward the ground. The exhaust system, including all fasteners, shall be constructed of corrosion resistant materials. The system shall be protected from damage which could result from traversing rough terrain.

3.4.1.2 Transmission. A commercial automatic transmission shall be provided, complete with a hydraulic torque converter (see 6.3.2). The transmission shall have a minimum of three forward speeds and shall have a torque capacity of at least 110 percent of the maximum net engine torque.

3.4.1.2.1 Transfer case. The truck shall be equipped with a two speed transfer case.

3.4.1.3 Axles. The load rating of each axle shall be not less than the imposed load on the axle, measured at the ground, with the vehicle at gross weight. The rear axle shall be equipped with limited slip or positive locking differential.

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3.4.1.4 Drive hubs. Unless a full time four wheel drive system with an interaxle compensation device is provided, the vehicle shall be equipped with automatic drive hubs for disengaging the front wheels from the powertrain.

3.4.2 Steering system. The vehicle shall be equipped with a power assisted steering system, with manual steering available in the event of power assist failure. The steering system shall not require more than five turns lock to lock.

3.4.3 Service brakes. The vehicle shall be equipped with power assisted brakes. Front brakes shall be of the disc type.

3.4.4 Frame. The truck frame shall be the chassis manufacturer's standard or optional assembly for this type of vehicle, suitable for the performance requirements specified herein. The contractor shall not weld to the frame assembly or add any holes to the frame rail flanges.

3.4.4.1 Bumpers. Bumpers shall be provided at the front and rear of the truck. The rear bumper shall be of the step type, recessed and reinforced for mounting a pintle hook in accordance with MS 51335; the pintle hook shall be furnished and mounted on the vehicle. The rear bumper shall have a minimum 12 inch step depth that spans the entire width of the vehicle. The front and rear bumpers shall be free from all sharp edges and shall have cross-cut or rounded corners to minimize the hazard to personnel walking around the vehicle.

3.4.4.2 Brush guard. A structural brush guard shall be securely mounted to the front of the vehicle to provide adequate protection against brush and other debris encountered during off road use. The brush guard shall protect, but not limit or obstruct, the operation of the winch, headlights and warning lights. The outer corners of the brush guard shall be rounded and shall incorporate other protection as required to minimize the hazard to personnel walking around the front of the vehicle.

3.4.4.3 Towing devices. Not less than two hooks, loops or pins for towing the vehicle shall be mounted at the front and rear of the vehicle.

3.4.4.4 Winch. A 12 volt, 8,000 pound capacity electric worm gear type winch assembly shall be mounted at the front of the truck. The winch shall be provided with 150 feet of 5/16 inch diameter, 7 x 19 aircraft cable, Type I, Composition B, in accordance with MIL-W-83420. Winch line pull rating on drum shall not be less than 8,000 pounds. The winch shall have forward and reverse drum drive and a drum clutch to permit free spooling of the cable. The winch shall be equipped with a four way roller guide, weatherproof cover, snatch block assembly and chocker chain. A clevis hook shall be installled at the end of the cable for quick attachment or removal of the snatch block assembly.

3.4.4.5 Suspension. The vehicle shall be equipped with a suspension system adequate to provide complete control, without sway, to the driver under all operating conditions specified herein.

3.4.5 Wheels and tires. The vehicle shall be equipped with wheels and tires which meet all Tire and Rim Association requirements for this application. Tubless steel belted radial ply tires with all terrain type tread shall be provided.

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3.4.5.1 Spare wheel. If specified (see 6.2), a spare wheel shall be furnished. The spare wheel shall be identical to the wheels mounted on the vehicle.

3.4.5.2 Spare wheel and tire assembly. If specified (see 6.2), a spare wheel and tire assembly, inflated to the chassis manufacturer's recommended pressure, shall be furnished. The spare wheel and tire assembly shall be identical to the wheel and tire assemblies mounted on the vehicle.

3.4.5.3 Tire tools. Unless otherwise specified (see 6.2), the vehicle shall be equipped with at least a jack, a jack handle and a wheel lug nut wrench for replacement of the mounted tire and wheel assembly with a spare assembly. The jack, at its closed height, shall fit under the chassis manufacturer's approved lift points and shall be capable, without blocking, of raising any wheel of the loaded vehicle to a height sufficient to allow removal and replacement of the wheel and tire assembly.

3.5 Cab. A two door full width cab shall be furnished. An interior dome light and map light, designed and located for use by personnel in any seating position, shall also be provided. A space at least 18 inches long by 12 inches wide by 12 inches high, shall be provided for mounting a radio for base operations. The location shall be such that radio controls shall be located on a two tier communications organizer. Tier units shall adjust from a minimum width of 5 1/4 inches to a maximum width of 8 7/8 inches, and shall have a height adjustment to allow adjusting from 3 5/8 inches minimum to a maximum height of 6 1/4 inches to accomodate different electronic communications equipment. The organizer shall also be provided with a quick disconnect to allow for quick removal/installation for servicing the equipment. A universal floor mount shall be provided to allow mounting on the center, left or right of the drive shaft housing or directly to any flat portion of the cab floor. The communications organizer shall be of all steel construction with a black finish. The organizer shall be furnished with each vehicle and located in a convenient location to both the driver and crew member. The cab shall have a minimum of three coat hooks located in the rear of the cab for storing protective clothing and helmets. A light shall be provided on the rear of the cab to provide a completely lighted area for the bed of the truck.

3.5.1 Seats. The cab shall be provided with two seats; a driver's seat and one crew member's seat. The driver's seat shall be a bucket type seat with an automatic locking , push button or lever actuated, forward folding seat back. With the seat folded forward, easy access shall be provided to the storage area of the extended cab. The crew member's seat shall be of tubluar steel construction with a padded vinyl seat, back rest, and head rest. The head rest shall be adjustable to allow full protection at all desired settings. The back rest shall be hinged so as to flip forward to permit fast and easy exit. The seat shall be provided with horizontal rotation capability to allow the seat to rotate 90 degrees towards the cab door to allow for unobstructed emergency entrance and exit. A hand operated locking device, operable by a fire fighter wearing arctic mittens, shall be provided to securely lock and release the seat. A walkway bracket designed to accomodate any breathing apparatus and allow for fast and easy removal shall be provided. Seat belts with retractors shall be provided at all seating positions.

3.5.2 Doors. Each door shall be furnished with a door restraint. The restraint shall prevent the door from opening past the 90 degree mark measured from the closed position. The doors in the open position shall not interfere with normal entering or exiting of the cab. Each door shall be furnished with an armrest.

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3.5.3 Windshield and windows. The vehicle's windshield and all cab windows shall have tinted glass.

3.5.4 Air conditioning. The vehicle shall be equipped with the chassis manufacturer's standard or optional air conditioning system.

3.5.5 Mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall be of the west coast type, having separate flat and convex areas enclosed in a common housing. The flat portion of each mirror shall have not less than 100 square inches of reflective area. The mirrors shall be positioned so that the driver can clearly observe the roadway and vehicles approaching from the rear on either side.

3.5.6 Accessories. The cab shall be equipped with all accessories furnished as standard equipment by the chassis manufacturer. The following minimum equipment shall be furnished:

a. Speedometer and odometer.

b. Tachometer.

c. Engine oil pressure gauge.

d. Engine coolant temperature gauge.

e. Voltmeter or ammeter.

f. Fuel level gauge.

g. Dual sun visors.

3.6 Utility body. The vehicle shall be equipped with an all aluminum utility body. The utility body shall be a unit that is commercially available or a unit that is produced by a recognized manufacturer of aluminum service bodies. The body shall be of unitized welded construction with a minimum thickness of 0.125 inches. The understructure shall be braced and reinforced at all critical stress points. The utility body shall have a minimum of two compartments and a maximum of three compartments on each side with sufficient space to store the below listed fire fighting equipment. The area above the compartments shall have aluminum treadplate to allow fire fighters to walk above the compartment areas. A drip rail shall be mounted over all compartment doors. The body shall be equipped with wheel well liners that allow clearance for the installation and removal of tire chains. A removable, heavy ribbed or diamond plate aluminum rub rail along each side of the lower body shall be provided. The utility body shall have sufficient floor area for the mounting of the extinguishing system and to allow access to the agent tanks from the rear of the vehicle. All components necessary for operating the extinguishing systems shall be centrally located and easily accessible to a fire fighter standing on the ground. When specified (see 6.2), the items listed shall be provided. Mounting clamps for securing the fire fighting equipment shall be provided with each vehicle. For the contractor first article test sample the fire fighting equipment shall be provided (see 4.4.2.1).

a. One Ametek aircraft skin penetrator, Model ASP/AA-2, stored in a compartment on the right side of the utility body.

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b. Air hose reel assembly (see 3.13.1).

c. One Mine Safety Appliance Company, Number 401. Self-Contained Breathing Apparatus (SCBA).

d. Two SCBA air bottles stored in one compartment.

e. Two 2 1/2 gallon capacity water pressured fire extinguisher, one stored in a compartment on each side of the utility body.

f. Two each rechargeable portable lanterns, one stored on each side of the vehicle. Each lantern shall have a minimum 25000 CP and minimum 300 hour battery use life. The lanterns shall have waterproof switches and a leather shoulder strap. g. One Pry-Axe forcible entry tool (see 3.16).

h. One Aircraft Tools Incorporated, Part Number AT501C, cutter tube and hose.

i. One set of vehicle chocks, snatch block assembly with chocker chain.

j. Two sets of protective clothing (turnout coat and pants, gloves, goggles, boots, and helmet with faceshield) , one stored in a compartment on each side of the utility body.

k. Servicing equipment and tools.

3.6.1 Compartments and shelving. Each compartment on opposite sides of the utility body shall be dimensionally equivalent. The compartments shall have weathertight seals with bottoms flush and level with, or slightly above, the bottoms of the doors. Compartment doors shall be of the upward acting, counter balance, roll-up type aluminum doors. The roll-up type door panels shall be designed with weatherproof seals between panels to allow for freeze-free operation in low temperature regions. The doors, when in the open position, shall provide an opening substantially the full length of the compartment. All fasteners shall be recessed or flush with the compartment panels. Each door shall be provided with an automatic device to hold them in the open position. In addition, handles and latches shall be provided for opening and closing the doors. Handles and latches shall be operable with one hand by an operator wearing arctic mittens. The front compartment shall have a full width adjustable and removable shelf mounted 26 inches minimum from the floor of the compartment. The shelf shall be adjustable at two inch increments up or down and shall be designed to support a load of at least 100 pounds.

3.7 Electrical system.

3.7.1 Alternator. The engine shall be equipped with an alternator of sufficient capacity to maintain all vehicle electrical components operating simultaneously. The system shall supply full power requirements for operation of the vehicle and all equipment and accessories at engine idle speed without system degradation. An engine high idle speed control, which automatically increases the engine speed, may be provided. The device shall be preset so that when activated, it will operate the engine at the appropriate speed; maximum engine speed shall not exceed the engine manufacturer's recommended speed for this type application. The design shall allow for disengagement when the transmission is engaged or the parking brake released. Minimum capacity of the alternator system shall be 130 amperes.

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3.7.2 Battery. Two 12 volt batteries, with a minimum rating of 550 cold cranking amperes, shall be furnished. The batteries shall comply with the maintenance free characteristics of W-B-131. A battery selector switch shall be installed, complete with battery isolator. The battery selector switch shall be easily accessible to the driver inside the cab. The battery selector switch shall comply with KKK-A-1822A, paragraph 3.7.7. It shall be provided with an oil and fuel resistant housing with weatherproof "O" ring seals at critical areas. A independent feed terminal to guard against accidental battery rundown shall also be provided. The battery selector switch shall be provided with heavy gauge copper electrical contacts and 3/8 inch diameter copper stud terminals with brass hex nuts for attaching cables.

3.7.3 Communication equipment circuit. A dedicated 12 volt electrical circuit with an individual circuit breaker shall be provided with wiring to the space provided for radio installation (see 3.5). The circuit shall be rated at 15 amperes minimum. The communications equipment circuit shall come equipped with adequate noise suppression devices to preclude interference from warning lights, heater motor, windshield wipers and other electrically operated equipment.

3.7.4 Lighting system.

3.7.4.1 Floodlights. Two each 200 watt telescopic quartz floodlights shall be installed behind the cab of the vehicle. Telescopic tubing shall be stainless steel or anodized aluminum. Floodlights shall be readily adjustable and shall lock securely in place. Telescoping capability shall be a minimum of two feet above the highest point of the truck with 360 degrees rotation capability. Floodlights shall also provide for vertical angle movement of 60 degrees above and 35 degrees below the horizontal. The adjustment devices shall be of the twist knurled knob type. Each floodlight shall be controlled by a lighted indicator switch located on the vehicle's instrument panel. Each switch shall be easily accessible to both the driver and crew member.

3.7.4.2 Spotlight. A standard automotive spotlight shall be installed on the top forward center location of the cab. The spotlight shall be adjustable from inside the cab and shall not interfere with normal driving operations. The spotlight shall be capable of 270 degrees horizontal rotation and a vertical angle movement of at least 60 degrees above and 35 degrees below the horizontal. Spotlight rotation shall not interfere with the light bar and speaker specified in 3.7.4.3. All moving parts of the spotlight assembly shall be lubricated and sealed.

3.7.4.3 Warning lights. Two red flashing lights shall be installed on the front grill and at the rear of the vehicle in a protected location. The warning lights shall be three inches in diameter minimum. Warning lights shall operate when the light bar specified in 3.7.4.4 is activated. When specifed (see 6.2) blue warning lights shall be provided.

3.7.4.4 Light bar. A removable light bar, with dual directional (front and rear) reflecting mirrors and two sealed beam synchronized rotating lamps shall be provided. The lamps shall be of the Halogen type with a total maximum beam candlepower of 60,000. The light bar shall be securely attached to the structural rain gutters of the cab; no holes shall be permitted in the roof of the cab. The light bar shall be controlled by a lighted indicator switch located on an overhead panel provided with the light bar or on the vehicles instrument panel. The light bar shall be equipped with a speaker contained in a stainless steel housing.

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3.7.4.5 Back up alarm and lights. An audible electric alarm shall be provided to caution personnel in the area when the vehicle is in reverse gear. The alarm shall be in accordance with SAE J994. In addition, the vehicle shall be equipped with adequate rear lighting for backing of the vehicle. The lighting shall be activated when the vehicle is engaged in reverse.

3.7.4.6 Siren. An electric siren, with basic siren tones (wail, yelp, hi-lo), TAP II, PA, radio broadcast and bulit-in air horn sound with siren override shall be provided for operating the light bar speaker (see 3.7.4.4). The siren system shall include a built-in front microphone with siren override switch for PA use, illuminated non-glare control panel with gain control, fuse for protection against failure, built-in radio interconnect wires. The siren shall have all necessary wiring and connections so that it is adaptable to two way radios which use either carbon or magnetic type microphones and so that it can be used for a public address system and for broadcasting incoming radio messages through the siren speaker. The electronic siren shall be solid-state, with integrated circuits and silicon output transistors. The siren shall have a 100 watt capacity.

3.7.4.7 Electrical component protection. All electrical components shall be so designed, protected and so located that their normal functioning will not be impaired by heavy rains, formation of condensation, pressure cleaning, spillage of extinguishing agents or leaks in the piping system.

3.8 Winterization system. When specified (see 6.2), the vehicle shall be equipped with a winterization system to permit storage and operation in ambient temperatures as low as -40 degrees F. Heaters shall operate on 110 volt alternating current (AC) and shall be wired through a junction block to a single three pronged (male) weatherproof slave receptacle for receiving external power and grounding the vehicle. When specified (see 6.2), heaters shall operate on a 220 volt system. A three wire connecting cable, 25 feet long and of adequate capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard weatherproof three pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77 (c)(7). Electrical insulation of connecting cable shall withstand normal operating stersses in low ambient air temperature without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for storage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

a. Coolant heater, 1500 watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. Engine thermostat, with an operating range of 170 degrees F to 195 degrees F, shall be installed.

b. Immersion type engine oil heater, 300 watt minimum rating, with 170 degrees F to 195 degrees F thermostat, shall be installed in the oil pan through any convenient opening. A dipstick type heater is not acceptable.

c. Battery heater shall be provided, having a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10 degrees F during vehicle exposure in ambient air temperatures as low as -40 degrees F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80 degrees F.

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3.9 Extinguishing system. The extinguishing system shall consist of the following major components.

DESCRIPTION	SEE REQUIREMENT
Halon 1211 agent system	3.10
Dry chemical agent system	3.11
Pressure system	3.12
Hose reel assembly	3.13
Air compressor	3.14
Skid mounted frame	3.15

3.9.1 Functional design. The extinguishing system shall be designed to operate efficiently when skid mounted in the bed of the utility body. The unit shall be designed to dispense both Halon 1211 and dry chemical agent either singly or simultaneously. The Halon 1211 system and dry chemical system shall be in accordance with Section 4-13.8 and 4-13.7 of NFPA 414. The extinguishing system shall be equipped with safety devices to adequately protect all hazards incident to their operation.

3.9.2 Agent system vlaves. Unless otherwise specified, agent system valves shall be of the quarter turn full-on to full-off type. All valves shall be mounted so as to be readily removable without distrubing piping and other components. All valves shall seal and be operable without damage under the low ambient temperatures and environmental conditions specified herein. All parts which could wear shall be replaceable or adjustable. Eech valve shall have a nameplate clearly identifying the valves function and open/closed positions.

3.9.3 Agent system piping. Piping in contact with the agent shall be brass. The piping shall incorporate seamless tubing and groove type couplings (threaded pipe and fittings) to the maximum practical extent. All piping sections between agent system components shall be readily removable without distrubing components. Large pipe sections, and pipe sections which could be subjected to excessive movement during operation; shall be firmly mounted or braced. Sharp bends and changes in cross sections shall be avoided.

3.9.4 Extinguishing system nameplates. A nameplate shall be mounted on each agent tank and shall contain the following information:

a. Type extinguishing system.

b. Capacity (pounds nominal).

c. Weight full (pounds nominal).

d. Weight empty (pounds).

e. Operating pressure (PSIG).

f. Hydrostatic test date (month-day-year).

3.10 Halon 1211 system. The vehicle shall be equipped with a 500 pound capacity bromochlorodifluromethane (Halon 1211) fire extinguisher system. The Halon 1211

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system shall be capable of extinguishing three 600 square foot liquid fuel fires in accordance with UL 711 without recharge. Halon 1211 shall be in accordance with MIL-B-38741. All components, except the agent tank, which come into contact with the agent shall be brass unless otherwise approved by the procuring activity.

3.10.1 Halon 1211 agent tank. The Halon 1211 agent tank, minimum 500 pound capacity, shall be spherical in shape or cylinderical with semi-ellipsoidal heads. The tank shall be designed and constructed in accordance with Section VIII of the ASME Boiler and Pressure Vessel Code or in accordance with Department of Transportation (DOT) 178.61. Operating pressure shall be 225 + 25 PSIG. The agent shall not occupy more than 85 percent of the agent tank.

3.10.2 Connectors and valves. The agent tank shall be equipped with a quick connect charging coupling, installed in a horizontal position. The coupling shall be of copper alloy material with a chromium surface treatment, with a maximum operating pressure of 300 psi, it shall also be self-sealing with spring push-pull for quick disconnect on all ends. It shall have a nominal thread size of .500 inches and a quick disconnect nominal size of 1.00 inch with NPT thread series. A shut off valve shall be located between the charging coupling and agent tank. The tank shall also be equipped with an agent discharge valve, hose blow-down valve, tank valve located in the nitrogen piping and a tank blow-down valve. The tank blow-down valve shall be vented beneath the bed of the vehicle. Valves shall be of the ball type and of brass construction with teflon seats and seals. The nitrogen supply line connection shall be equipped with a check valve to prevent flow of agent into the pressure regulator. Valve controls for pressurizing the agent tank and charging the handline shall be operable from the hose reel end of the unit. Valves shall not be easily subject to mechanical, chemical or other damage. All controls shall be located in an easily accessible location for quick operation. Lifting provisions shall be provided to enable removal with the agent tank, without damage.

3.10.3 Gauges. The agent tank shall be equipped with a pressure gauge and an agent level gauge. The gauges shall have hermetically sealed shatterproof dails and shall be located on the side of the Halon 1211 agent tank or in the system piping. Gauges shall be properly marked to indicate both the proper operating zone and danger zone of the Halon extinguishing system. Gauges shall be easily visible and accessible to an operator standing on the ground.

3.10.4 Recharge hose assembly. A 20 foot Halon 1211 charging hose assembly, Ansul Company Part Number 403541, or equal, shall be furnished with each unit.

3.10.5 Purge system. A purge system shall be provided such that pressurized gas can be routed directly to the inlet side of the handline, by-passing the agent tank for purging the handline of residual Halon 1211.

3.10.6 Nitrogen pressure system. The nitrogen pressure system shall provide pressure for Halon discharge to meet or exceed the performance and purge system requirements specified herein. The system shall include the nitrogen cylinder and its mounting, nitrogen cylinder quick release valve, pressure regulator and necessary piping and hoses (see 3.12). The nitrogen cylinder charging connection shall conform to MS 28889.

3.10.7 Piping and hoses. All piping and hoses used in the Halon 1211 system, including the nitrogen pressure system, shall have a burst pressure of not less than

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four times the maximum operating pressure. Piping and hoses shall be as specified in paragraph 3.9.3 and shall be in accordance with UL 92.

3.10.8 Leakage. The Halon 1211 agent system components shall not exceed the following maximum leakage rates.

AREA	MAXIMUM LEAKAGE RATE (OUNCES/YEAR)
Welded or soldered seams (each seam)	0.6
Threaded connections (each connection)	5.0
Valves (each valve)	5.0
Gauges (each gauge)	5.0

3.10.9 Pressure regulator. A pressure regulator shall be provided to reduce the supply pressure to the Halon 1211 system operating pressure. The regulator shall be designed for an inlet pressure of up to 3,360 PSIG and shall be permanently set to deliver an outlet pressure of 225 (plus or minus 25 PSIG) to the agent system within 10 seconds of opening the nitrogen quick release valve. The regulator shall be equipped with a relief valve capable of handling the full gas flow of the regulator; the relief valve shall relieve at 325 PSIG (plus or minus 25 PSIG). The pressure regulator setting shall be adjusted and sealed in a manner to discourage manipulation in the field.

3.11 Dry chemical system. The vehicle shall be equipped with a minimum 400 pound capacity potassium bicarbonate base dry chemical extinguisher system. Potassium bicarbonate base dry chemical shall conform to O-D-1407. The hoseline and nozzle shall have a static wire of sufficient size to ground the unit and prevent shocks.

3.11.1 Dry chemical agent tank. The dry chemical agent tank shall be designed and constructed in accordance with Section VIII of the ASME Boiler and Pressure Vessel Code or in accordance with Department of Transportation (DOT) 178.61. The agent tank shall be spherical in shape or cylindrical with semi-ellipsoidal heads. Operating pressure shall be 225 PSIG (plus or minus 25 PSIG). The agent tank shall be provided with a readily accessible agent fill opening of not less than four inches in diameter. The opening shall provide for unobstructed filling of the tank to its rated capacity of uncompacted agent in accordance with 3.11.

3.11.2 Fill cap. The fill opening shall be provided with a threaded cap incorporating a safety vent hole for pressure venting while at least 3 1/2 threads are still engaged. The cap shall incorporate long handles (wings) to assist in its removal. The cap shall have an adequately protected tamper proof visual indicator to indicate when the tank is pressurized. A funnel, with strainer screen, shall be provided to assist in filling the agent from standard pails.

3.11.3 Dry chemical piping and hoses. All piping and hoses used in the dry chemical system, including the nitrogen pressure system, shall have a burst pressure of not less than four times the maximum operating pressure. Unless otherwise specified herein, piping design, hoses and fittings shall be in accordance with UL 92 and 3.9.3.

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3.11.4 Connectors and valves. The agent tank shall be equipped with an agent discharge valve, a tank blow-down valve, hose blow-down valve and a tank valve located in the nitrogen piping. The tank blow-down valve shall be vented beneath the bed of the vehicle. These shall be ball valves of brass construction with teflon seats and seals. The nitrogen supply line connection shall be equipped with a check valve to prevent flow of agent into the pressure regulator. Valves shall not be easily subject to mechanical, chemical or other damage. All controls shall be located in an easily accessible location for quick operation. Valve controls for pressurizing the agent tank and charging the handline shall be operable from the hose reel end of the unit. Lifting provisions shall be provided to enable removal with the agent tank without damage.

3.11.5 Pressure gauge. The dry chemical agent tank shall be equipped with a hermetically sealed pressure gauge. The pressure gauge shall be located on the side of the dry chemical agent tank or in the system's piping. The gauge shall be easily visible and accessible to an operator standing on the ground.

3.11.6 Purge system. A purge system shall be provided such that pressurized gas can be routed directly to the inlet side of the handline, by-passing the agent tank for purging the handline of residual dry chemical.

3.11.7 Nitrogen pressure system. The nitrogen pressure system shall provide pressure for the dry chemical discharge to meet or exceed the performance and purge system requirements specified herein. The system shall include the nitrogen cylinder and its mounting, nitrogen cylinder quick release valve, pressure regulator and necessary piping and hoses (see 3.12). The nitrogen cylinder charging connection shall conform to MS 28889.

3.11.8 Pressure regulator. A pressure regulator shall be provided to reduce the supply pressure to the dry chemical system operating pressure. The regulator for the dry chemical system shall be equivalent to the regulator specified in 3.10.9. The pressure regulator setting shall be adjusted and sealed in a manner to discourage manipulation in the field.

3.12 Pressure system.

3.12.1 Nitrogen cylinders. Two nitrogen cylinders, one per each extinguishing agent, shall be furnished. The pressure system shall be of sufficient capacity to properly fluidize dry chemical agent, pressurize and expel each agent at rated delivery and to purge the handlines of residual agents. Two DOT 3AA-2400, 300 cubic foot, compressed gas cylinders shall be provided, installed on rollers in a horizontal position for ease of replacement. The method of securing the nitrogen cylinders shall hold the cylinders firmly and safely in place while withstanding all shocks and vibrations induced during vehicle operations. Cylinder mounting shall permit cylinder replacement by two operator in less than five minumtes.

3.12.2 Nitrogen cylinder valves and gauges. A UL listed nitrogen cylinder valve shall be provided attached to the standard valve on the nitrogen cylinder. The valve shall be of the quick opening lever actuated type. The vlave shall permit instantaneous full gas flow from the cylinder by pivoting the valve lever to a maximum 90 degrees. An anti-tamper device shall secure the valve when in the closed position. The valve outlet shall conform to the Compressed Gas Association CGA 580 valve outlet and connection standard for nitrogen service. All moving parts shall be

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hardened stainless steel, except for the camshaft. Also provided shall be an integral pressure relief device and pressure gauge. The pressure relief device shall limit pressure to 3360 - 4000 PSIG. The pressure gauge shall be sealed with a shatterproof gauge face cover.

3.12.3 High pressure hose. A flexible high pressure hose shall be used between each nitrogen valve and each pressure regulator to permit easy replacement of the nitrogen cylinder. The hose shall be equipped with one fixed coupling and one swivel adapter. The hose assembly shall be service tested at nominal gas pressure of 2,400 PSIG and shall withstand hydrostatic test at a pressure of 4,000 PSI for one minute with no movement of the couplings nor permissible leakage. To permit diffusion of trapped gases, the hose cover shall be pin pricked with two rows of holes approximately 180 degrees apart, with the holes spaced 5/8 to 3/4 inch apart, parallel to the axis, 1/32 inch deeper than the cover, for a minimum of 32 effective holes per foot.

3.12.4 Low pressure hose. A flexible low pressure hose shall be used between the pressure regulator and the agent tank gas tube fitting. The hose shall be equipped with one fixed coupling and one swivel fitting. The hose assembly shall be capable of withstanding 400 PSI for five minutes without movement of the couplings. To permit diffusion of trapped gases, the hose cover shall be pin pricked with two rows of holes approximately 180 degrees apart, with the holes spaced 5/8 to 3/4 inch apart, parallel to the axis, 1/32 inch deeper than the cover, for a minimum of 32 effective holes per foot.

3.13 Hose reel assembly. Dual electric rewind hose reel assemblies consisting of a reel, hoseline, and an agent nozzle shall be furnished and mounted on the skid frame at the rear of the vehicle. The mounting on the skid (see 3.15) shall be as low as possible to the floor of the utility body. The hose reel assemblies shall be easily accessible and fully operable by a fire fighter standing on the ground to the rear of the vehicle.

3.13.1 Hose reels. Each hose reel shall incorporate an electric motor rewind and a manual rewind for use in the event of electric rewind failure. The electric motor rewind shall be capable of an average rewind speed of at least 75 feet per minute. An adjustable friction device to prevent overtravel during handline withdrawal operations shall be provided. Horizontal and vertical rollers shall be provided to reduce chaffing of the hose. The hose reels shall have 90 degree swing joint inlets. The reels shall have a capacity of a minimum of 150 feet of one inch inside diameter hoseline. Operation shall permit easy one man withdrawal and rewind of the handline.

3.13.2 Hoseline and couplings. The hose reel specified in 3.13.1 shall be equipped with 150 feet of one inch inside diameter twin hoseline and couplings in accordance with UL 92 and MIL-C-7580. Hoses shall be capable of withstanding the pressures as specified in 3.10.7 and 3.11.3.

3.13.3 Handline nozzle. The handline nozzle shall be constructed of brass, bronze or anodized aluminimum. The nozzle shall have a single pistol grip handle with finger grooves and a horseshoe type operating valve handle. The nozzles shall provide the required nonaspirating flow rates and ranges for both agents without assistance from downwind conditions. Swivel connections shall be provided between the nozzle and handline to provide for 360 degree adjustment. The Halon 1211 nozzle shall be designed so an aircraft skin penetrator (see 3.6 a) can be easily inserted

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and secured for agent discharge. A quick release type holder for the handline nozzle shall be provided and installed in an easily accessible location to a fire fighter standing on the ground at the rear of the vehicle. Quick release type holders shall provide easy attachment and removal by an operator wearing arctic mittens.

3.14 Air compressor. A portable or engine driven air compressor shall be provided. The air compressor system should include, but not be limited to an air compressor, evaporator, air dryer and storage tank. The compressor shall be capable of delivering a sufficient amount of compressed air to adequately and continuously operate the aircraft skin penetrator (see 3.6). A pressure gauge with a minimum diameter of one inch shall also be provided and located in an easily visible location. The compressor shall allow for operation with the air hose reel (see 3.14.1).

3.14.1 Air hose reel assembly. An automatic or electric rewind air hose reel assembly for operation of the aircraft skin penetrator (see 3.6 a) shall be provided and securely mounted on the right side of the vehicle in a compartment. The hose reel shall consist of 150 feet minimum of 1/4 inch inside diameter hose on a suitable reel assembly. Burst pressure of the hose shall be not less than four times the operating pressure. The reel assembly shall have the capability to automatically rewind the hose when not in use and also provide manual rewind capabilities. The hose shall be provided with quick disconnect couplings and a shutoff valve.

3.15 Skid frame. The skid frame shall be constructed of steel or aluminum. Each connecting cross member shall have a continuous weld on the inside or outside cross members; spot welding is unacceptable. The frame shall consist of two main channel side members with a minimum of five internal connecting channel cross members; each agent tank shall rest on a channel cross member. For installation, the skid frame shall be easily and firmly bolted to the bed of the utility body and shall be designed so no reinforcements of the truck body are neccessary. Rustproof ball transfer/roller assemblies or non-rotating, non-ferrous, anti-friction nylon disks of at least 1 1/2 inches in diameter shall be provided to allow easy removal of the nitrogen cylinders. A minimum of four sets of ball transfers/rollers or disks shall be provided for each cylinder. The skid frame shall provide for installation of the items specified in 3.10, 3.11, 3.12, and 3.13. The entire unit shall be attached to withstand the jars, vibrations, shocks, etc, which the unit will experience during operations. When mounted to the utility body, all major components shall be positioned so they can be easily and quickly serviced and removed when neccessary.

3.15.1 Installation. The attachment of the extinguishing system to the bed of the vehicle shall require no more than two hours by one person. The installation shall not require the use of special tools; only standard handtools may be used. When mounted in the pickup bed, all agent system controls shall be readily accessible to operating personnel standing on the ground, at the rear of the vehicle.

3.15.2 Hoisting provisions. The skid mounted fire extinguishing system shall have hoisting provisions at four locations. Hoisting provisions shall be so located that the unit can be safely hoisted by crane with cable and hook attachment without causing damage to the components.

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3.15.3 Forklift provisions. The skid mounted fire extinguishing system shall be designed to allow lifting of the unit with a standard forklift truck from either side or at the rear of the skid.

3.16 Fire fighting equipment. The following items shall be furnished. Each item shall be securely stowed in a convenient location with quick release clamps operable by a fire fighter wearing fireman's gloves or arctic mittens.

a. One pry-axe multi-purpose forcible entry cutting tool with prick point, pry prong, leather sheath and instruction manual stored in the right rear compartment of the utility body (see 3.6).

b. One each 14 foot two section aluminum extension ladder in accordance with NFPA 1931. The ladder shall be mounted on the top right side of the utility body and shall be readily accessible to a fire fighter standing on the ground. When in the stowed position, the ladder shall not interfere with any normal operating procedures. c. One each hand portable dry chemical fire extinguisher for Class D type fires with a minimum UL rating for six pounds of magnesium chips.

3.17 Finish and protective coatings.

3.17.1 Cleaning, painting, plating, anodic films and chemical treatments. Cleaning, painting, plating, anodic films and chemical treatments shall be in accordance with MIL-STD-808. Unless otherwise approved by the procuring activity, primer shall be in accordance with MIL-P-23377.

3.17.2 Finish.

3.17.2.1 Lime yellow. Unless otherwise specified (see 6.2), the outer surface of the vehicle shall be painted with DuPont Imron polyurethane paint, or equal (see 6.3), Lime Yellow, Color Number 13670 of FED-STD-595. Chrome items on the vehicle shall not be painted Lime Yellow. The front and rear bumpers of the vehicle shall be chrome, diamond plate or matte black.

3.17.2.1.1 Equality tests:

a. Tests in accordance with MIL-C-46168:

(1) Paragraph 4.3.17 Flexibility.
(2) Paragraph 4.3.23 Accelerated weathering.

b. Tests in accordance with MIL-C-83286:

(1) Paragraph 4.8.6 Low temperature resistance.
(2) Paragraph 4.8.10 Resistance to taping.
(3) Paragraph 4.8.12 Pencil hardness.

3.17.2.2 Forest green. When specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments (except the cab) shall be painted with polyurethane paint in accordance with MIL-C-83286 or shall be DuPont Imron, or equal, Forest Green, Color Number 24052 of FED-STD-595. All exterior trim items (such as wheels, wheel covers, metal radiator grilles, bumpers, headlight trim rings, door handles and drip rails) shall match the exterior finish of the vehicle.

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3.17.2.3 Desert sand. When specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments (except the cab) shall be painted with polyurethane paint in accordance with MIL-C-46168 or MIL-C-53039, Desert Sand, Color Number 30310 of FED-STD-595. All exterior trim items (such as wheels, wheel covers, metal radiator grilles, bumpers, headlight rings, door handles, and drip rails) shall be painted with the same paint as the exterior of the vehicle.

3.17.3 Rustproofing. Unless otherwise specified (see 6.2), the vehicle shall be rustproofed in accordance with the contractor's standard commercial practice. The first article units shall not be rustproofed until after approval of the first article test report.

3.18 Markings. Unless otherwise specified, markings for vehicles painted Lime Yellow shall be applied with reflectorized tape in accordance with L-S-300, Type I, Class 3, Reflectivity 1, white on a red background. Markings for vehicles painted Forest Green shall be applied with black paint, Color Number 37030 of FED-STD-595, and markings for vehicles painted Desert Sand shall be applied with white paint, Color Number 37875 of FED-STD-595.

3.18.1 Identification markings. Unless otherwise specified (see 6.2), the vehicle shall be marked for the appropriate service in accordance with MIL-STD-1223.

3.19 Identification plates and operating instructions. All identification plates, instruction plates, data plates, etc, shall conform to MIL-P-514 and be of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12.

3.19.1 Identification plate. An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the vehicle in a readily accessible location. The identification plate shall contain the following information:

NOMENCLATURE

MANUFACTURER'S MODEL NUMBER

MANUFACTURER'S SERIAL NUMBER

MANUFACTURER'S NAME

MANUFACTURER'S ADDRESS

MANUFACTURER'S SERVICE PHONE NUMBER

DATE OF MANUFACTURE

CONTRACT NUMBER

NATIONAL STOCK NUMBER

3.19.2 Control markings. All controls, valves, gauges and indicators used in the operation of the vehicle shall be identified by securely attached nameplates.

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3.19.3 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the appropriate controls. The instructions shall be clear, concise and adequate to enable operation of the vehicle without damage to the equipment or injury to personnel and shall refer to the components as identified by nameplates.

3.19.4 Lubrication plate. A lubrication plate shall be provided directing attention to all lubrication fittings and components which require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures.

3.19.5 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

a. Side and rear silhouette views of the vehicle.
b. Horizontal and vertical location of the center-of-gravity of the vehicle in transport condition.
c. Shipping weight.
d. Loading cubage.
e. Overall length, width and height.
f. Tiedown instructions for air transport in Types C-130, C-141 and C-5 Aircraft.

3.20 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.21 Welders and welding.

3.21.1 Welders. All welders employed in the fabrication of the vehicle, except for the commercial chassis, shall be certified in accordance with the applicable requirements of MIL-STD-1595, the American Welding Society, or the American Society of Mechanical Engineers.

3.21.2 Welding. The surface parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof and service loadings.

3.22 Screws, rivets, gears and levers.

3.22.1 Screw assemblies. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.22.2 Riveting. Riveting operations shall be performed to ensure tight fit and complete heading of the rivets.

3.22.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. When required for accurate adjustment, gear assemblies shall be free of backlash.

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3.22.4 Cleaning. The vehicle shall be thoroughly cleaned. Loose, spattered or excess solder, welding slag, stray bolts, nuts and washers, rust metal particles, pipe compound and other foreign matter shall be removed during and after final assembly.

3.22.5. Workmanship. The vehicle, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring and riveting, marking of parts and assemblies; alignment of parts and tightness of assembly fasteners; etc.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use any inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government. The Government reserves the right to perform any if the inspections set forth in this specification where such inspections are deemed necessary to assure that the vehicle conforms to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractors overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of kown defective materials, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

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c. Time finished.

d. Ambient temperature.

e. Engine speed.

f. Engine oil pressure.

g. Engine coolant temperature.

h. Transmission oil temperature.

i. Vehicle odometer reading at the start and finish of each test.

j. Engine hourmeter reading at the start and finish of each test.

4.3.3 Preparation for tests. The vehicle, equipped with all specified auxiliary equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.3.4 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any liquid, including fuel, coolant, lubrication or hydraulic fluid, under any operating condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking and while traversing rough terrain.

h. Conditions which present a safety hazard to personnel during operation, service or maintenance.

i. Overheating of the engine, transmission or any other vehicle component.

4.4 First article testing.

4.4.1 First article test samples. The first article test samples shall be preproduction items which meet the requirements of this specification. The sample shall be identical to the items procured under the terms of the contract. Approval

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of the first article samples shall not constitute a waiver by the Government of its rights under the provisions of the contract. Two first article samples shall be produced and tested as follows:

a. Contractor Test Sample (see 4.4.2.1)

b. User Test Sample (see 4.4.2.2).

4.4.2 First article tests.

4.4.2.1 Contractor test sample. The contractor test sample shall be subjected to the test described in 4.6.1 through 4.6.13 (see 6.2.2).

4.4.2.2 User test sample. The user test sample shall be subjected to the tests in 4.6.1, 4.6.4.2, 4.6.4.3, 4.6.4.4, 4.6.5, 4.6.6, 4.6.7, 4.6.9.5, 4.6.10.4, 4.6.10.5, 4.6.10.6 and 4.6.11.5.

4.4.3 Contractor test equipment. The contractor's test sample shall be equipped with all equipment referenced herein for first article testing.

4.5 Quality conformance testing. Quality conformance testing shall be classified as follows:

a. Individual Testing (see 4.5.1).

b. Random Sample Testing (see 4.5.2).

4.5.1 Individual testing. Each vehicle shall be subjected to the tests specified in 4.6.1, 4.6.10.4 and 4.6.10.5.

4.5.2 Random sample testing. Any production vehicle may be selected by the procuring activity at any time to be subjected to the tests described in 4.6.3.1, 4.6.4, 4.6.5, 4.6.7, 4.6.10 and 4.6.11.5.

4.5.2.1 Rejection and retest. When a vehicle fails to meet the requirements of this specification, vehicles to be delivered pursuant to the contract may not be accepted until the cause of the failure is determined, corrective action is taken by the contractor and the component that failed is successfully retested to assure compliance with this specification. Tests not involving the failed component may be continued pending the investigation of a test failure.

4.6 Test methods.

4.6.1 Examination of product. The vehicle shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and their application, welding and riveting, and markings. For the first article and random sample vehicles, this examination shall be accomplished by use of a checklist which lists each requirement not validated by a test and the results of the examination.

4.6.2 Agent system pressure test.

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4.6.2.1 Hydrostatic pressure. The agent tanks and all components, except pressure relief valves, shall be subjected to a hydrostatic pressure test in accordance with Section VIII of ASME Boiler and Pressure Code for Unfired Pressure Vessels, for a service pressure of 250 PSI. In addition, each hose assembly shall be hydrostatically presure tested at not less than two times the maximum operating pressure for at least 15 minutes. At the conclusion of the test, the test fluid shall be completely drained and the components completely dried.

4.6.2.2 Relief valve test. Each pressure relief valve in the extinguishing system shall be subjected to a gradually increasing pressure of clean dry oil free air or nitrogen until it starts to discharge. The pressure shall then be reduced until the valve completely reseals. This procedure shall be repeated twice. The valve shall then be checked for leakage with its maximum working pressure applied. Failure to open or reseal within the specified limits or indication of leakage shall be cause for rejection.

4.6.3 Preliminary operation.

4.6.3.1 Road test. The vehicle, with extinguishing system filled to operating capacity and the nitrogen bottles fully charged, shall be operated over the following types of roads for the distances specified. The test fluid used for the vehicle's engine shall be Grade JP-8 Turbine Fuel in accordance with MIL-T-83133 or JET-A1 Fuel.

a. The vehicle shall be operated over hard surface highways for a distance of 300 miles at an average speed of 40 MPH. During the test, the following operations shall be performed:

   (1) Normal or average stops, ten times.

   (2) Quick stops from a speed of at least 30 MPH by instantaneous application of the brakes, five times.

   (3) Sharp right turns and sharp left turns, five times each.

b. The vehicle shall operate over graded gravel roads for a distance of 300 miles at an average speed of 30 MPH.

c. The vehicle shall operate over a Belgian Block course for a distance of 300 miles at an average speed of 10 MPH.

d. The vehicle shall be tested to demonstrate compliance with 3.3.1(1).

4.6.4 Halon 1211 system tests.

4.6.4.1 Halon 1211 leakage test. The agent tanks shall be charged with a minimum of 50 pounds of Halon 1211 or refrigerance 12. The agent tank, and all components which come into contact with Halon, shall then be pressurized with nitrogen to 250 PSIG minimum and soaked at an ambient temperature of not less than 68 degrees for four hours minimum. At the conclusion of the soak period, a Halogen leak detector, such as the GE H-25, shall be used to determine the leakage rate at each seam, valv and gauge. Failure to comply with 3.10.8 shall be cause for rejection.

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4.6.4.2 Halon 1211 agent discharge. Using the recharge hose assembly specified in 3.10.4, the agent tank shall be filled to its rated capacity of 500 pounds and pressurized by the nitrogen cylinder to 225 (plus or minus 25 PSI). The handline nozzel shall be held parallel to the ground by one operator. The agent shall then be discharged from the handline nozzle at the maximum steady discharge rate. This rate shall be maintained until pressure pulsations or flow interruptions occur. Discharge duration, range, pattern and precentage of agent discharge prior to pressure pulsations or flow interruptions shall be and observed for compliance with operational requirements. In addition, it shall be demonstrated that the handline can be purged of residual Halon 1211 utilizing the pressurized nitrogen cylinder to the inlet side of the handline.

4.6.4.3 Aircraft skin penetrator test. The aircraft skin penetrator (see 3.6) shall be removed from its stored position. The Halon 1211 nozzle shall then be inserted and secured to the skin penertator. The agent shall be discharged through the penetrator and initiated at a minimum of five PPS. Upon completion, the quick disconnect capability shall be demonstrated and the penetrator shall be returned to its stored position. This test may be combined with 4.6.6.

4.6.4.4 Fire test. The Halon 1211 system shall extinguish three 600 square foot liquid fuel pan fires in accordance with UL 711. The handline shall then be purged of residual Halon 1211. Neither Halon nor nitrogen shall be added to the system between fires.

4.6.4.5 Halon 1211 rate of flow test. The agent tank shall be filled to its rated capacity of 500 pounds and pressurized by the nitrogen cylinder to 225 (plus or minus 25 PSI). The agent shall then be discharged through the handline nozzle for a period of two-thirds (60 seconds) the effective duration. The rate of flow shall be calculated. Failure to comply with 3.3.2 (a) shall be cause for rejection. A suitable reciever may be used to capture the discharged Halon. A suitable reciever is defined as a storage container which will enable uncontaminated recovery and storage of agent for re-use.

4.6.5 Dry chemical system tests.

4.6.5.1 Dry chemical agent fill test. Utilizing the funnel specified in 3.11.2 and standard pails, the dry chemical agent tank shall be filled to its rated capacity of 400 pounds. Agent capacity shall be measured with the dry chemical in fresh fill uncompacted condition.

4.6.5.2 Discharge of dry chemical agent. The dry chemical agent tank shall be pressurized by the nitrogen cylinder to 225 PSI (plus or minus 25 psi). The handline nozzle shall be held parallel to the ground by one operator. The agent shall then be discharged from the handline nozzle at the maximum steady discharge rate. This rate shall be maintained until pressure pulsations or flow interruptions occur. Discharge rate, range, pattern and percentage of agent discharged prior to pressure pulsations or flow interruptions shall be observed for compliance with 3.3.3. It shall be demonstrated the handline can be purged of residual dry chemical utilizing the pressurized nitrogen cylinder to the inlet side of the handline.

4.6.5.3 Dry chemical fire test. Perform the test specified in 4.6.4.4, except use dry chemical to extinguish the fires. The handline shall then be purged of residual dry chemical. Neither dry chemical agent nor nitrogen shall be added to the system between fires.

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4.6.6 Air compressor and hose assembly test. The aircraft skin penetrator (see 3.6 a) shall be removed from its stored position and the hose (see 3.14.1) attached and secured to both the penetrator and air compressor (see 3.14). It shall be demonstrated the air compressor can deliver a sufficient amount of air to adequately and continuously operate the penetrator. In addition, the quick disconnect capability of the air hose to the penetrator and air compressor shall be demonstrated. Upon completion, each item shall be returned to its stowed position. Provisions shall be provided to measure the pressure and flow rate at several points in the system.

4.6.7 Alternator output test. With the vehicle stationary and operating at its normal operational speed, the air conditioner and all electrical lighting equipment (floodlights, spotlights, headlights, warning lights, light bar and rear cab light) shall be simultaneously operated at maximum equipment output. Operation, as described, shall be maintained for a minimum of one hour. Any evidence of battery discharge during the test shall be cause for rejection. It shall also be demonstrated that the system is capable of operating and maintaing the winch (see 3.4.4.4) under its maximum pull without system degradation.

4.6.8 Hoisting and lifting test. The extinguishing system shall be raised and lowered into the truck by a forklift and/or crane. There shall be no interference between the pickup bed, extinguishing system, skid frame or prime lifting vehicle. Each item shall be inspected for damage as a result of these operations. The following shall be tested empty and when filled to rated capacity:

a. Dry chemical agent tank (crane).

b. Halon 1211 agent tank (crane).

c. Skid mounted extinguishing system (crane and forklift).

4.6.9 Mobility tests. The vehicle shall perform the following tests (see 4.6.9.1 thru 4.6.9.7.2) with Grade JP-8 Turbine Fuel in accordance with MIL-T-83133 or JET-A1 fuel as the engines test fluid.

4.6.9.1 Maximum speed and acceleration. The time-speed-distance relationship of the fully loaded vehicle shall be determined while accelerating from a standing start to maximum speed on a dry level paved surface to demonstarte compliance with 3.3.1 (a) and 3.3.1 (b). Not less than three runs, each in opposite directions, shall be conducted. Maximum speed and acceleration shall be the average for all runs.

4.6.9.2 Grade operation. The fully loaded vehicle shall be driven up and down a 50 percent dry paved grade. The maximum speed that can be maintained while climbing the grade shall be measured. Engine oil pressure shall be observed and shall not be less than the manufactruer's recommended minimum pressure at the engine operating speed. Engine starting ability shall be demonstrated in both directions on the grade.

4.6.9.3 Side slope operation. The fully loaded vehicle shall be driven in both directions on a side slope of 20 percent. Engine oil pressure shall be observed ar shall not be less than the engine manufacturer's recommended pressure at the engin operating speed. Engine starting ability shall be demonstrated in both directions on the slope.

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4.6.9.4 Ramp test. The fully loaded vehicle shall be driven forward from a hard flat horizontal surface and ascend a ramp with a minimum length of 1.2 times the vehicle wheelbase and a slope of 30.8 percent. It shall than proceed accross the ramp crest onto an upper connecting hard horizontal surface until all tires are resting on the upper surface, down the ramp, and onto the lower horizontal surface. The test shall then be repeated in the opposite direction.

4.6.9.5 Road handling test. The fully loaded vehicle shall be driven on a dry level paved surface to demonstrate compliance with 3.3.1 (m). The vehicle shall operate through right and left turns and maneuver through a standard slalom type course. A minimum of ten right turns, ten left turns and five slalom passes shall be conducted at an average speed of no less than 40 MPH. Throughout the road handling test, the vehicle shall be observed for its roadability, ease of handling and tendency to sway or yaw.

4.6.9.6 Suspension articulation. The vehicle shall be tested to demonstrate compliance with 3.3.1 (k). The sides of the blocks may be sloped.

4.6.9.7 Brake tests.

4.6.9.7.1 Service brake tests. The vehicle shall be tested in accordance with SAE J786a to demonstrate compliance with 3.3.1 (i).

4.6.9.7.2 Parking brake tests. The vehicle shall be tested in accordance with SAE J360 to demonstrate compliance with 3.3.1 (j).

4.6.10 Design verification.

4.6.10.1 Weight and weight distribution. The weight on each corner of the vehicle shall be measured with the unit standing on a flat level surface, both in the empty and the fully loaded conditions. The fully loaded vehicle shall include ballast to simulate a 250 pound operator in each seating position.

4.6.10.2 Electromagnetic interference measurement. The vehicle and all electrical components shall be tested in accordance with MIL-STD-462 to demonstrate compliance with 3.2.6.

4.6.10.3 Sound level testing. The vehicle shall be tested in accordance with SAE J336a and J366b to demonstrate compliance with 3.2.7.

4.6.10.4 Hose reel test. Each hose reel shall be unwound and rewound to demonstrate compliance with 3.13.1. There shall be no portion of the hose that interferes with or does not fit on the reel. Each hose reel shall be checked for smooth operation when the hose is being unwound and rewound by one operator where applicable, the following operations shall be conducted on the hose reel as installed on the truck:

a. electrical operation.

b. Mechanical operation.

c. Rewind force.

d. Rewind rate.

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e. Unwind force.

4.6.10.5 Electrical lighting test. Floodlights and spotlights shall be tested to demonstrate conformance to 3.7.4.1 and 3.7.4.2. Each light shall be tested for its ease of accessibility, adjustment, and rotation. In addition, it shall be demonstrated the rear cab light specified in 3.5 provides a completely lighted area for the bed of the truck.

4.6.10.6 Equipment storage test. Equipment requiring secure stowage or attachment to the vehicle shall be verified for ease of attachment/removal by a fire fighter wearing arctic mittens or fireman's gloves. Unobstructed accessibility to controls, gauges, valves and components shall also be verified. Special attention shall be given to the following:

a. Removal and replacement of each nitrogen cylinder within five minutes. Each nitrogen cylinder shall be recharged to its operating capacity.

b. Attachment of the extinguishing system to the vehicle by one operator within two hours. Installation shall be accomplished in accordance with the instructions provided.

c. Removal, replacement and accessibility of the forcible entry tool specified in 3.16.

d. Removal, replacement and accessibility of the aircraft skin penetrator and SCBA (see 3.6)

e. Verification each cab door does not open over 90 degrees from the closed position.

f. Removal, replacement, and accessibility of the aluminum extension ladder (see 3.16.).

4.6.11 Environmental tests.

4.6.11.1 Low temperature storage and operation. The complete vehicle, including all accessories, shall be tested in accordance with Test Method 502.2, Procedures I and II, of MIL-STD-810 to demonstrate compliance with 3.2.3 (a). Temperature sensor locations shall be approved by the procuring activity. Operational checkouts shall be performed in accordance with paragraph 4.6.11.5. Engine starting aids may be used (see 3.4.1.1.2). When applicable, a fine mist of water shall be applied to the entire windshield to form a thin film of ice. The defroster shall clear the windshield within 15 minutes and maintain clear vision throughout the remainder of the test. "Standard ambient" and "controlled ambient" shall be defined as the ambient condition of the test site. For Procedure I, the vehicle shall be exposed to the minimum required cold storage temperature of 3.2.3 (a) in a steady-state condition for 24 hours. For Procedure II, the vehicle shall be exposed to the maximum required hot operational temperature of 3.2.3 (a) in a steady-state condition.

4.6.11.2 High temperature storage and operation. The complete vehicle, including all accessories, shall be tested in accordeance with Test Method 501.2, Procedures and II, of MIL-STD-810 to demonstrate compliance with 3.2.3 (a). Temperature sensor

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locations shall be approved by the procuring activity. Operational checkouts shall be performed in accordance with 4.6.11.5. 'Standard ambient' and 'controlled ambient' shall be defined as the ambient condition of the test site. For Procedure I, the vehicle shall be exposed to the maximum required hot storage temperature of 3.2.3 (a) in a steady-state condition for 24 hours. For Procedure II, the vehicle shall be exposed to the maximum required hot operational temperature of 3.2.3 (a) in a steady-state condition.

4.6.11.3 Rain. The complete vehicle, including all accessories, shall be tested in accordance with Test Method 506.2, Procedure III, of MIL-STD-810 to demonstrate compliance with 3.2.3 (c). Operational checkouts shall be in accordance with 4.6.11.5. The interior of the cab and all compartments shall be examined for water; any evidence of leakage shall be cause for rejection.

4.6.11.4 Other environmental tests. The components and materials used in the construction of the vehicle shall be subjected to, or certified as having passed, or being capable of passing, the following test methods of MIL-STD-810 to demonstrate compliance with 3.2.3:

a. Humidity - Method 507.2, Procedure I.

b. Salt Fog - Method 509.2, Procedure I.

c. Sand and Dust - Method 510.2, Procedures I and II. For Procedure I, air velocity shall be 1750 ft/min and high temperature adjustment shall be 125 degrees F. For Procedure II, air velocity shall be 5700 ft/min; sand concentration of 0.0623 g/ft3, plus or minus 0.015 g/ft3; 90 minutes per face duration.

d. Freezing rain/icing - Method 521.0, Procedure I.

4.6.11.5 Operational checkout. All controls, levers, mechanisms and doors shall be actuated a sufficient number of times to ensure consistency of operation and to determine whether the test exposure produced system degradation. These actions are to result in the operation of the following systems, but will not limit to:

a. Engine starting, operation and accessories; accessory powerplants.

b. Cab climatic systems, including defroster.

c. Cab doors and windows.

d. Lights (i.e. headlamps, turn signals, gauges/controls, work lights, cab lights, etc.) e. Brakes, stationary (i.e. service, parking and emergency).

f. Windshield wipers.

g. Drivetrain (distance limited by chamber size).

h. Steering.

i. Winch.

j. Hose/electrical lines/connectors.

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k. Equipment/compartment covers/doors.

l. Miscellaneous equipment.

4.6.12 Maintainability demonstration. Compliance with the maintenance downtime requirements shall be demonstrated by task performance. All scheduled maintenance tasks and all tasks listed in 3.2.3.2 shall be performed. In addition, corrective maintenance tasks shall be selected by the procuring activity from an approved list of tasks provided by the contractor in the first article test procedure. Not less than 25 percent or more than 50 percent of the total number of tasks shall be demonstrated to verify an average maintenance downtime within the required limit. As part of this demonstration, the recommended frequencies of the scheduled maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of scheduled maintenance downtime per measure of use, such as calendar time, hours of operation or miles traveled. The capability of performing maintenance tasks by personnel wearing arctic mittens shall be demonstrated.

4.6.13 Air transportability analysis. A detailed engineering analysis shall be conducted to demonstrate compliance with 3.2.8. The analysis shall include the tiedowns and all major components and their ability to withstand the accelerations specified in MIL-STD-1791. The evaluation shall also include a dimensional analysis for the vehicle while loaded and unloaded aboard a type C-130 Aircraft.

4.6.14 User test. The vehicle shall be delivered (where specified in the contract or purchase order, see 6.2) for 30 days of use. The test period shall begin after the vehicle has been inspected and prepared for use, and shall be for up to 240 hours of operation. The vehicle shall be available for any eight hour period specified by the Government each day, seven days a week. Any period of more than two hours that the vehicle is out of service for maintenance or repair shall not be considered part of the test period. The vehicle shall be operated, and scheduled maintenance shall be performed by Government personnel using the operation and maintenance procedures contained in the preliminary technical data to be supplied with the vehicle. Any parts or corrective actions required shall be provided by the contractor. Test operations will simulate normal military usage and will not exceed any operating parameters defined by this specification. The sole criteria for acceptance/rejection of the user test shall be based on the performance requirements of this specification. During each day of operation, up to 10 pan fires and simulated aircraft fires may be extinguished. The vehicle may be driven up to 200 miles per day, over hard surfaces, gravel roads and cross country terrain. Sudden stops, minimum radius turns, maximum speed and maximum acceleration may be included in the daily operation. The adequacy of vehicle lighting shall be verified during night operation. All systems and subsystems of the vehicle may be tested at the beginning and end of the test period.

4.15 Refurbishment. After successful completion of all testing, each vehicle shall be restored to like new condition (see 6.3.4).

5. PACKAGING.

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with the terms of the contract.

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6. NOTES

6.1 Intended use. The vehicle is intended for fire fighting at flightlines and other airfield facilities.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Spare wheel, if required (see 3.4.5.1).

c. Spare wheel and tire assembly, if required (see 3.4.5.2).

d. Firefighting equipment, if required (see 3.6).

e. Tire tools, if required (see 3.4.5.3).

f. Blue warning lights, if required (see 3.7.4.3).

g. Winterization system, if required (see 3.8).

h. Winterization system, 220 volt, if required (see 3.8).

i. Finish color required, if not specified, Lime Yellow, Color Number 13670 of FED-STD-595 (see 3.17.2).

j. Rustproofing, if other than the manufacturer's standard commerical practice is required (see 3.17.3).

k. Identification markings required (see 3.18.1).

l. Markings required (see 3.18).

m. Destination of delivery to perform user test.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporate a DD Form 1423, Contract data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

	Paragraph no.	Data requirement title	Applicable DID No.
a.	4.5.2, 4.6.2.1, 4.6.4.2, 4.6.5.2, 4.6.9.2, 4.6.14	First Article Inspection Report	DI-T-4902

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b. 3.1, 4.4.2.1 First Article Inspection Procedure DI-T-4901

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer).

6.2.3 First article. When a first article inspection is required, the items should be selected from preproduction or production vehicles. The first article should consist of two units. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract (see 3.1 and 4.4.1)

6.3 Definitions.

Belgian Block.	A Belgian Block course is paved with unevenly laid granite or basalt block, forming an undulating surface. It duplicates a rough cobblestone road such as is found in many parts of the world. A cobblestone course may be used as an alternate.
Commerical.	A manufacturer's standard product, having been marketed, in production and sold in substantial quantities to the general public for a minimum of one year prior to the date of invitation for bid or request for proposal. An item is 'sold in substantial quantities' only when the quantities regularly sold are sufficient to constitute a real commerical market. Nominal quantities, such as models, samples, prototypes or experimental units, do not meet this requirement. The 'general public' is a significant number of buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification, items acquired for 'Government end use' include items acquired for foreign military sales.
Cross country terrain.	Cross country terrain consists of open fields, broken ground and uneven terrain in general.
Equal.	Having performance identical to or superior to DuPont Imvon when subjected to specified tests
Like new condition.	Like new condition refers to the restoring of the unit to its original condition and appearance. Any part or component of the vehicle, including tires and finish, with appreciable wear incurred during first article testing shall be replaced or reconditioned.

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6.4 Subject term (key word) listing.

Firefighting, truck
Airfield ramp, firefighting, truck
Air transportable, airfireld ramp, firefighting, truck
Diesel engine driven, firefighting, truck
Four wheel drive, diesel engine driven, firefighting, truck
Twin agent, firefighting, truck
Dry chemical, airfield ramp, firefighting, truck
Halon 1211, airfield ramp, firefighting, truck
Flightline, diesel engine driven, firefighting, truck
Facilities, airfield, twin agent, firefighting, truck
Skid mounted, dry chemical, airfield ramp, firefighting, truck
Utility bidy, twin agent, diesel engine, firefiighting, truck

6.5 Changes from previous issue. Asterisks (or verticle lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Air Force-99

Preparing activity:
Air Force-84

Agent:
Air Force-99
(Project 4210-F230)

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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